Aquablend — MILP Mathematical Formulation

Sprint 1 Network formulation, applied to the toy model

MILP Stream

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1 Overview

Water corporations draw drinking water from multiple sources, such as reservoirs, river intakes, and groundwater bores, that differ in cost and in raw water quality (pH, alkalinity, turbidity, and so on). Before that water can be supplied for human consumption, it is treated at a plant and delivered to demand zones. The question this model answers is: which sources should be drawn from, how much from each, and how should that water be routed to plants and zones, so that demand is met, and water quality stays within safe limits, at the lowest possible cost?

This is formulated as a Mixed-Integer Linear Program (MILP). Whether a source, plant, or transfer link is used at all is a binary decision; how much water is drawn or flows along each link is a continuous decision. The objective balances fixed activation costs against the variable cost of drawing and treating water.

Treatment cost here is a flat rate per megalitre (ML) processed, and water quality only enters as a feasibility requirement: a blend must stay within limits, but blending sources more favourably does not itself lower the treatment cost. Modelling that difference, where a better blend reduces treatment cost, is a future enhancement.

The toy case works with the simplest version of this problem: one treatment plant and one demand zone, fed by several candidate sources. With only one plant and one zone, there is no routing choice to make: every source that's switched on sends its full draw to the plant, and the plant sends everything on to the zone. As a result, the model reduces to deciding which sources to activate and how much to draw from each. Section 7 below derives this toy case as a special case of the general network formulation, so the same model scales up to multiple plants and zones later without being rebuilt from scratch.

2 Notation

Convention. This is a single-period model (one representative day), so no day index is used. Parameters are uppercase; decision variables are lowercase, using Greek letters for binary decisions and lowercase Latin for continuous volumes. A subscript is an index; an overbar/underbar is an upper/lower bound.

2.1 Sets

Notation	Description
$\mathcal{S}$	set of water sources, $s \in \mathcal{S}$
$\mathcal{T}$	set of treatment plants, $t \in \mathcal{T}$
$\mathcal{Z}$	set of demand zones, $z \in \mathcal{Z}$
$\mathcal{P}$	set of water quality parameters, $p \in \mathcal{P}$

2.2 Parameters

Notation	Description	Units
D_z	volume of water to deliver to demand zone z	ML/day
F_s	fixed cost of activating source s	\$/day
F_t	fixed cost of activating treatment plant t	\$/day
C_s	unit cost of drawing water from source s	\$/ML
C_t	unit treatment cost at plant t	\$/ML
$\underline{W}_s$	minimum daily withdrawal from source s	ML/day
$\overline{W}_s$	maximum daily withdrawal from source s	ML/day
$\underline{V}_t$	minimum daily throughput of plant t	ML/day
$\overline{V}_t$	maximum daily throughput of plant t	ML/day
$\overline{L}_{st}$	maximum flow on the link from source s to plant t	ML/day
$\overline{L}_{tz}$	maximum flow on the link from plant t to zone z	ML/day
Q_{sp}	value of quality parameter p in source s	units of p
$\underline{Q}_p$	lower regulatory limit for parameter p	units of p
$\overline{Q}_p$	upper regulatory limit for parameter p	units of p

Costs and limits are treated as fixed. The units of a quality parameter depend on p (e.g. mg/L, NTU). Water quality parameters that are not linear in their natural units are stored in Q_{sp} as a transformed value that blends approximately linearly.

2.3 Decision variables

Notation	Domain	Description	Units
α_s	$\{0,1\}$	1 if source s is activated	—
a_s	≥ 0	volume drawn from source s	ML/day
β_t	$\{0,1\}$	1 if plant t is activated	—
γ_{st}	$\{0,1\}$	1 if source s sends water to plant t	—
b_{st}	≥ 0	volume of water from source s to plant t	ML/day
δ_{tz}	$\{0,1\}$	1 if plant t sends water to demand zone z	—
c_{tz}	≥ 0	volume of water from plant t to demand zone z	ML/day

3 Example network structure

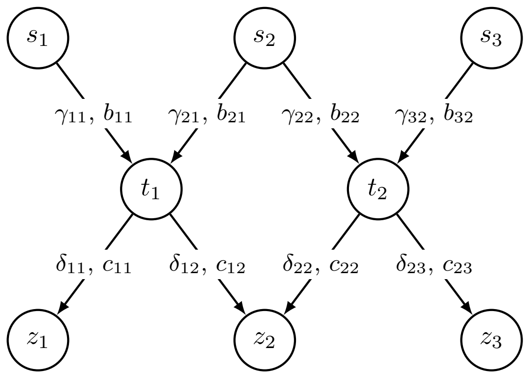

Figure 1: One example of the network structure (sources $\rightarrow$ treatment plants $\rightarrow$ demand zones). Each source-plant arc carries (γ_{st}, b_{st}) and each plant-zone arc carries (δ_{tz}, c_{tz}). The formulation is generic over any number of sources, plants, zones and the arcs connecting them; this is only one instance.

4 Objective

Minimise total cost over a single period: the fixed cost of activating each source and plant, the per-ML cost of drawing water, and the per-ML cost of treating it (charged on the volume each plant processes, $\sum_s b_{st}$).

$$\min \quad \underbrace{\sum_{s\in\mathcal{S}} F_s\alpha_s}_{\text{source activation}} \quad + \quad \underbrace{\sum_{t\in\mathcal{T}} F_t\beta_t}_{\text{plant activation}} \quad + \quad \underbrace{\sum_{s\in\mathcal{S}} C_s a_s}_{\text{drawing water}} \quad + \underbrace{\sum_{t\in\mathcal{T}} C_t \sum_{s\in\mathcal{S}} b_{st}}_{\text{plant treatment}} . \tag{1}$$

5 Constraints

Non-negativity. All continuous flow variables must be non-negative.

$$a_s, b_{st}, c_{tz} \geq 0, \qquad \forall s \in \mathcal{S}, t \in \mathcal{T}, z \in \mathcal{Z}. \tag{2}$$

Demand satisfaction. The total volume of water sent from all treatment plants to demand zone z via the links c_{tz} must be at least the demand D_z for that zone.

$$\sum_{t\in\mathcal{T}} c_{tz} \geq D_z, \qquad \forall z \in \mathcal{Z}. \tag{3}$$

Oversupply Flexibility The demand constraint uses $\geq$ rather than equality ($=$) to allow operational flexibility. In principle, this permits the solution to exceed demand. However, cost minimization naturally prevents oversupply: the optimizer has no incentive to draw more water than needed, as additional volume incurs both variable and treatment costs.

Oversupply may occur only if quality constraints require blending from multiple sources to meet regulatory limits. For the toy scenario (demand 1200 ML/day, maximum available capacity 1680 ML/day), the model produces exactly demand at minimum cost with zero excess.

This formulation provides flexibility for future extensions. A multi-period model with storage capacity could leverage this structure to allow demand smoothing across time periods.

Source capacity and activation. Water can only be drawn from a source if it has been activated. If source s is inactive ($\alpha_s = 0$), the draw a_s is forced to zero. If source s is active ($\alpha_s = 1$), the draw must lie between its minimum daily withdrawal $\underline{W}_s$ and maximum daily withdrawal $\overline{W}_s$.

$$\underline{W}_s\alpha_s \leq a_s \leq \overline{W}_s\alpha_s, \quad \forall s \in \mathcal{S}. \tag{4}$$

Plant capacity and activation. A plant can process water only if it has been activated. If plant t is inactive ($\beta_t = 0$), the total volume arriving from all sources is forced to zero. If plant t is active ($\beta_t = 1$), the total incoming volume must lie between its minimum daily throughput $\underline{V}_t$ and maximum daily throughput $\overline{V}_t$.

$$\underline{V}_t\beta_t \leq \sum_{s\in\mathcal{S}} b_{st} \leq \overline{V}_t\beta_t, \quad \forall t \in \mathcal{T}. \tag{5}$$

Plant flow conservation. The total volume leaving plant t toward all zones via the links c_{tz} equals the total volume entering it from all sources via the links b_{st}.

$$\sum_{z\in\mathcal{Z}} c_{tz} = \sum_{s\in\mathcal{S}} b_{st}, \quad \forall t \in \mathcal{T}. \tag{6}$$

Source flow conservation. The volume drawn from each source a_s must equal the total volume it sends to all plants via the links b_{st}.

$$a_s = \sum_{t\in\mathcal{T}} b_{st}, \quad \forall s \in \mathcal{S}. \tag{7}$$

Link capacity. Each link has a maximum flow and can carry water only if that link is active. For a source-to-plant link, the flow b_{st} is capped at $\overline{L}_{st}$ and forced to zero unless the link is active ($\gamma_{st} = 1$).

$$b_{st} \leq \overline{L}_{st}\gamma_{st}, \quad \forall s \in \mathcal{S}, t \in \mathcal{T}. \tag{8}$$

Similarly, a plant-to-zone link's flow c_{tz} is capped at $\overline{L}_{tz}$ and is zero unless the link is active ($\delta_{tz} = 1$).

$$c_{tz} \leq \overline{L}_{tz}\delta_{tz}, \quad \forall t \in \mathcal{T}, z \in \mathcal{Z}. \tag{9}$$

Links require active nodes. Links can only be active if the corresponding upstream node (source or treatment plant) is active. This is enforced by ensuring that the binary variables representing the links are less than or equal to the binary variables representing the activation of the sources and plants. A source-to-plant link requires the source to be active:

$$\gamma_{st} \le \alpha_s, \quad \forall s \in \mathcal{S}, t \in \mathcal{T}. \tag{10}$$

A plant-to-zone link requires the plant to be active:

$$\delta_{tz} \le \beta_t, \quad \forall t \in \mathcal{T}, z \in \mathcal{Z}. \tag{11}$$

Optional: a source-to-plant link could also require the plant to be active,

$$\gamma_{st} \le \beta_t, \quad \forall s \in \mathcal{S},\ t \in \mathcal{T}. \tag{12}$$

This is not explicitly needed, since zero flow to an inactive plant is already guaranteed by the plant capacity constraint (5). Suppose the plant t is inactive, so $\beta_t = 0$. Then (5) becomes

$$\sum_{s \in \mathcal{S}} b_{st} \le \overline{V}_t \, \beta_t = 0,$$

and since every flow is non-negative ($b_{st} \ge 0$), this forces $b_{st} = 0$ for all $s \in \mathcal{S}$.

Water quality. The water arriving at each plant must stay within the quality limits. For parameter p, $\sum_s Q_{sp} b_{st}$ is the total amount of p entering plant t and $\sum_s b_{st}$ is the total volume, so this keeps the average value of p at the plant between $\underline{Q}_p$ and $\overline{Q}_p$.

$$\underline{Q}_p \sum_{s \in \mathcal{S}} b_{st} \le \sum_{s \in \mathcal{S}} Q_{sp} b_{st} \le \overline{Q}_p \sum_{s \in \mathcal{S}} b_{st}, \quad \forall t \in \mathcal{T},\ \forall p \in \mathcal{P}. \tag{13}$$

Each Q_{sp} must be a quantity that blends linearly by volume so that summing each source's contribution, $\sum_{s \in \mathcal{S}} Q_{sp} \, b_{st}$, correctly represents the blended quality at the plant.

6 Water quality formulation logic

This section explains how the generic water quality constraint (13) is interpreted for turbidity, alkalinity, and pH. The purpose is to keep the quality formulation consistent with the existing parameter notation while preserving the linear structure of the MILP.

6.1 Interpretation of the generic quality parameter

In the general formulation, Q_{sp} represents the value of quality parameter p in source s. The set $\mathcal{P}$ may include parameters such as turbidity, alkalinity, and pH. For quality parameters that blend approximately linearly by volume, the source value can be used directly in Q_{sp}.

The constraint (13) is written in load form. It is equivalent to requiring the volume-weighted average quality at each treatment plant to remain between the lower and upper bounds for each quality parameter, but it avoids division by a decision variable.

6.2 Turbidity

Turbidity measures the cloudiness of water and is associated with suspended particles such as sediment, organic matter, algae, clay, or runoff material. In water supply modelling, turbidity is important because high turbidity may increase treatment requirements and affect disinfection performance [6, 7, 8].

For turbidity, the quality parameter is represented by setting $p = \text{turb}$. Thus, $Q_{s,\text{turb}}$ denotes the turbidity value of source s, while $\underline{Q}_{\text{turb}}$ and $\overline{Q}_{\text{turb}}$ denote the lower and upper turbidity limits.

The turbidity requirement is therefore handled directly by the generic water quality constraint (13) with $p = \text{turb}$. This avoids redefining the same load form constraint using separate turbidity specific equations.

6.3 Alkalinity

Alkalinity measures the ability of water to neutralise acid [3, 4, 9]. It represents the buffering capacity of water and is relevant to pH stability, corrosion control, treatment performance, remineralisation, and water stability in distribution systems [3, 9].

For alkalinity, the quality parameter is represented by setting $p = \text{alk}$. Thus, $Q_{s,\text{alk}}$ denotes the alkalinity value of source s, while $\underline{Q}_{\text{alk}}$ and $\overline{Q}_{\text{alk}}$ denote the lower and upper alkalinity limits.

The alkalinity requirement is therefore handled directly by the generic water quality constraint (13) with $p = \text{alk}$. This avoids redefining the same load form constraint using separate alkalinity specific equations.

6.4 pH transformation

pH is handled differently from turbidity and alkalinity because pH is a logarithmic measure of hydrogen ion concentration. Raw pH values are not averaged directly because direct pH averaging can produce a scientifically misleading blended value.

pH is defined as [5]:

$$\text{pH} = -\log_{10}([\text{H}^+]), \tag{14}$$

where $[\text{H}^+]$ represents hydrogen ion concentration [1].

To keep the MILP linear, pH is handled as a preprocessing transformation [5]. For the pH quality parameter, $Q_{s,\text{pH}}$ stores the transformed hydrogen ion concentration obtained from the raw source pH value:

$$Q_{s,\text{pH}} = 10^{-\text{pH}_s}. \tag{15}$$

The symbols $\underline{\text{pH}}$ and $\overline{\text{pH}}$ represent the raw regulatory lower and upper limits in pH units. These raw limits are converted once during data preparation before being passed into the MILP. If the acceptable raw pH range is:

$$\underline{\text{pH}} \le \text{pH} \le \overline{\text{pH}}, \tag{16}$$

then the corresponding transformed bounds stored for the pH quality parameter are:

$$\underline{Q}_{\text{pH}} = 10^{-\overline{\text{pH}}}, \qquad \overline{Q}_{\text{pH}} = 10^{-\underline{\text{pH}}}. \tag{17}$$

The direction of the bounds changes because pH and hydrogen ion concentration are inversely related [2]. A higher pH corresponds to a lower hydrogen ion concentration, while a lower pH corresponds to a higher hydrogen ion concentration.

After this preprocessing step, pH is handled by the same general water quality constraint (13), with $p = \text{pH}$. This avoids introducing a separate pH specific constraint in the MILP.

6.5 Blended pH recovery after optimisation

The optimisation model constrains pH through the transformed value $Q_{s,\text{pH}}$. After the solver finds the optimal source to plant flows, the blended hydrogen ion concentration at plant t is calculated as:

$$Q_{t,\text{pH}}^{\text{blend}} = \frac{\sum_{s\in\mathcal{S}} Q_{s,\text{pH}} b_{st}}{\sum_{s\in\mathcal{S}} b_{st}}. \tag{18}$$

The blended pH is then recovered in post-processing as:

$$\text{pH}_t^{\text{blend}} = -\log_{10}\left(Q_{t,\text{pH}}^{\text{blend}}\right). \tag{19}$$

This recovered pH value is not an MILP decision variable. It is a post-solve reporting value used for validation, interpretation, dashboard display, and output generation.

6.6 Linearity assumptions for water quality

The water quality formulation relies on simplifying assumptions that preserve linearity in the Sprint 1 MILP. The key assumptions are:

(i) Water from selected sources is completely mixed before quality is assessed.

(ii) The model represents a single time period, so temporal variation in water quality is not modelled.

(iii) Source quality values are fixed input parameters for the representative period.

(iv) Alkalinity and turbidity are approximated as volume-weighted linear blends.

(v) Raw pH values are not averaged directly; pH is transformed before optimisation.

(vi) No nonlinear chemical reactions between blended sources are modelled.

(vii) Water quality is checked at each treatment plant, using the blend of its incoming sources.

These assumptions allow the water quality constraints to remain linear and suitable for inclusion in the MILP formulation.

7 Application to the toy model

This network model is a generalisation of the Sprint 1 toy model, not a different problem: the toy is the special case with a *single* treatment plant and a *single* demand zone. With one plant and one zone, there is nothing to route: every activated source sends its whole draw to the one plant ($b_{st} = a_s$) and all of it reaches the one zone ($c_{tz} = \sum_s a_s$), so the routing variables γ, δ, b, c are fully determined ($\gamma = \delta = 1$, $b_{st} = a_s$, $c_{tz} = \sum_s a_s$) and only the source decisions α_s, a_s remain free. Formally this is the network model with $\mathcal{T} = \{t\}$, $\mathcal{Z} = \{z\}$, the plant fixed active ($\beta_t = 1$) and the routing fixed on ($\gamma_{st} = \delta_{tz} = 1$). Substituting these into the network equations gives the toy model formulation below.

7.1 Decision variables

Notation	Domain	Description	Units
α_s	$\{0, 1\}$	1 if source s is activated	—
a_s	≥ 0	volume drawn from source s	ML/day

7.2 Objective

Minimise the fixed activation cost of the chosen sources plus the per-ML cost of drawing and treating the water. The single plant is always on, so its fixed cost F_t is a constant and is dropped.

$$\min \quad \underbrace{\sum_{s \in \mathcal{S}} F_s \alpha_s}_{\text{source activation}} + \underbrace{\sum_{s \in \mathcal{S}} C_s a_s}_{\text{drawing water}} + \underbrace{C_t \sum_{s \in \mathcal{S}} a_s}_{\text{plant treatment}} . \tag{20}$$

7.3 Constraints

Demand satisfaction. The total volume drawn, all of which is delivered to the one zone, meets the zone's demand D_z.

$$\sum_{s \in \mathcal{S}} a_s \geq D_z. \tag{21}$$

Source capacity and activation. Identical to constraint (4).

Plant throughput. If plant minimum and maximum throughput limits are retained in the toy model, the fixed active plant must satisfy:

$$\underline{V}_t \leq \sum_{s \in \mathcal{S}} a_s \leq \overline{V}_t. \tag{22}$$

If the Sprint 1 toy model assumes that the single plant has sufficient feasible capacity for the tested scenarios, this constraint may be omitted from the simplified implementation.

Water quality. The blend of all drawn sources must lie within each parameter's limits.

$$\underline{Q}_p \sum_{s\in\mathcal{S}} a_s \le \sum_{s\in\mathcal{S}} Q_{sp} a_s \le \overline{Q}_p \sum_{s\in\mathcal{S}} a_s, \quad \forall p \in \mathcal{P}. \tag{23}$$

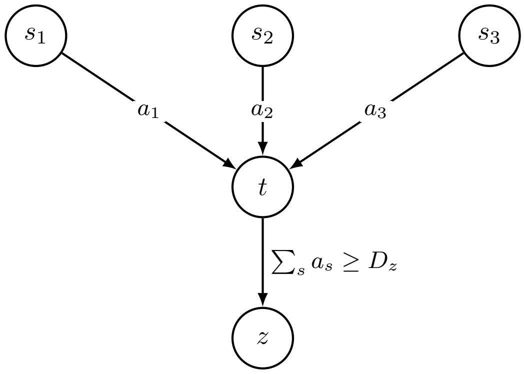

Figure 2: Toy model: three sources s_1, s_2, s_3 feed one plant t, which delivers to one zone z. Routing is fixed, so the delivered blend $\sum_s a_s$ must meet demand D_z and the quality limits.

Since this is just the network model with routing fixed, going the other way needs nothing new: free the routing variables and add more sources, plants and zones. The toy can therefore be built and tested first, then scaled up without reformulating.

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